MID TERM EXAMINATION 2014-15

University Roll No.....

B. Tech. II Year IV Semester

CSE-224: Computer Networks

Time: 01 ½ Hours **Total Marks: 20**

Note: Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.

Section-A

Note: Attempt all Questions **(1X5=5)**

Q1. What is the first address of a block of classless addresses if one of the addresses is 12.2.2.127/28?

Q2. An organization is granted a block of classless addresses with the starting address 199.34.32.0/27. How many addresses are granted?

Q3. If 5 is the number of bits for the sequence number; find the efficiency of Go-Back N and Selective repeat protocol, for transmission time of 1ms and propagation time of 1.5ms.

Q4. Explain the concept of piggybacking.

Q5. Draw the flow diagram of checksum error detecting technique.

Section-B

Note: Attempt any three questions **(2x3=6)**

Q1. An organization is granted the block 21.17.18.10/25. The administrator wants to create 3 subnets each of size 64, 32 and 32. Find the subnet mask. Also find the first address and last address of each subnet.

Q2. A bit stream 10011101 is transmitted using standard CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string

transmitted. Suppose the third bit from left is inverted during transmission. Show that this error is detected at receiver's end.

Q3.Suppose the four bit data word 1101 is to be encoded using an even-parity Hamming code. What is the binary value after encoding? Now assume that the 6th bit got corrupted during transmission. Find the error correcting codes.

Q4.Why the sender window size in Go-Back N ARQ is less than 2^m. Explain with a suitable example.

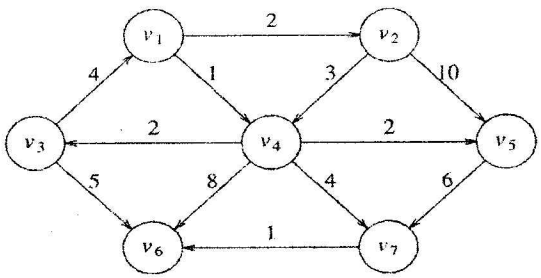
Section-C

Note: Attempt any three questions **(3x3=9)**

Q1.Consider a network with six routers A, B, C, D, E and F. Distance vector routing is used to route the packets. Suppose the following routers have sent the distance vectors to router B: from C(6,1,9,13,6,2),from D(17,13,6,10,11) and F(4,3,8,9,13).The measured distance to C,D and F are 5,3 and 4.What is B's new routing table.

Q2.Write the rules for creating a CIDR block. For a destination address 20.27.7.130 and the subnet mask 255.255.128.0.What is the subnet address and how many host per subnet are possible.

Q3.Explain the concept of link state routing. Apply Dijkastra's Algorithm on the given network having seven routers. Find the routing table for router v_1?



Q4. Explain IPv4 header in detail.

Printed Pages-2 **University Roll No.....**

B. Tech. II Year IV Semester, Second Term Examination, 2014-15

CSE-222: Introduction to Microprocessor

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions **(1*5=5)**

I. Which instruction is used to add 16 bit data in 8085 µp? Write an example.

II. Enlist the names of memory segments of 8086 µp.

III. What do you understand by debugging in 8086 µp?

IV. Write the names of two real time application, which are using concept of delay?

V. Define physical address in 8086 µp?

Section-B

Note: Attempt any three questions **(2*3=6)**

I. Is 8086 known as parallel processor? Justify your answer with diagram.

II. Calculate the value of counter (in Hexadecimal) for delay 0.2 second by using register pair, assuming the system clock frequency is 2MHz.

III. What do you mean by stack and subroutine? Discuss CALL and RET instruction with example.

IV. Calculate the delay (in ms) in the following loop, assuming the system clock period is 0.33 μs:

Label	Mnemonics		8085 T-States
	LXI	B, 12FFH	10
DELAY:	DCX	B	6
	XTHL		16
	NOP		4
	NOP		4
	MOV	A, C	4
	ORA	B	4
	JNZ	DELAY	10/7

Section-C

Note: Attempt any three questions **(3*3=9)**

I. How many flag bits are used in flag register of 8086? Explain controlling flags and overflow flag of 8086 μp with example.

II. Write assembly language program for multiply a byte data with a byte of data for 8085 μp. Also write description of each instruction.

III. Enlist addressing modes of 8086? Explain any four addressing modes of 8086 μp with example.

IV. Specify the number of times the following loops are executed. Why? Also calculate the total T-state for each. (Assume CY=0, Z=1, S=1 at initial position).

i)
```
      MVI  A, 17H
LOOP: ORA  A
      RAL
      JNC  LOOP
```

ii)
```
      MVI  A, 17H
LOOP: RAL
      ORA  A
      JNC  LOOP
```

iii)
```
      LXI  B, FF1FH
LOOP: DCR  B
      JNZ  LOOP
```

Printed Pages-2 University Roll No.....

B. Tech. II Year IV Semester, Mid Term Examination, 2014-15

CSE-223: Computer Graphics

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions **(1*5=5)**

I. Write the transformation matrix for mirror reflection with respect to the xy plane.

II. Write a 3D-transformation matrix for rotation about Y-axis.

III. Find the projected point for a point P (3, 4, 5) after applying orthographic projection on a plane Z = 0.

IV. Write the name of that parallel projection in which shadow on earth happens at 12 O'clock (i.e. Sun is at the highest point).

V. A point (7, 9) is reflected with respect to a line x = -y. Write the coordinate of reflected point.

Section-B

Note: Attempt any three questions **(2*3=6)**

I. Perform 60^0 rotation of a triangle A(0,0), B(1,1) and C(-1,-1) about a point P(0,1).

II. Prove that two 2D-rotations about origin commute i.e. $R_1R_2=R_2R_1$.

III. Find the transformation matrix for: cavalier projection with $\theta = 45°$ and for cabinet projection with $\theta = 30°$

IV. A unit cube is scaled uniformly by ½. Find the new coordinates of the scaled cube.

Section-C

Note: Attempt any three questions **(3*3=9)**

I. Explain Bezier curves with example. Write the properties of Bezier curves (At least four).

II. An object is rotated 45^0 about line L that has the direction V=j+k and passing through the point A(0,1,0). Find the composite matrix for the same.

III. Find the general form of matrix for reflection about a line L with slope m and y intercept (0, b).

IV. Enlist the significance of continuity conditions. Discuss parametric continuity conditions and differentiate it with geometric continuity conditions.

Mid Term Examination, 2014-15
Course: B.Tech (CSE), Year: II, Semester: IV
Operating Systems (CSE 221)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Differentiate between deadlock and starvation.
II. Define Race condition.
III. Write primitive of Test & Set instruction.
IV. Name different deadlock handling methods.
V. Differentiate between independent processes & Cooperating processes.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain Producer Consumer Problem and write the function for it with bounded buffer using shared memory solution.
II. Consider the scenario that in a system 7 processes named A, B, C, D, E, F, and G and 6 resources named R, S, T, U, V & W are present. Process A hold resource R and requesting for resource S, Process B is requesting for resource T, Process C is requesting for resource S, Process D hold resource U and requesting for resource S & T, Process E hold resource T

and requesting for resource V, Process F hold resource F and requesting for resource S and Process G hold resource V and requesting for resource U. Draw RAG for the above Scenario and convert it into WFG.

III. What is Semaphore? Explain different types of semaphores.

IV. Explain how deadlock can be prevented by ensuring that atleast one of the necessary conditions cannot hold.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. What is Critical Section problem? Explain Peterson's Solution to Critical Section Problem and state how it satisfies all the three requirements.

II. Assume that there are three resources A, B, and C and 4 processes P0 to P3. At T0 we have the following snapshot of the system:

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P_0	1	0	1	2	1	1	2	1	1
P_1	2	1	2	5	4	4			
P_2	3	0	0	3	1	1			
P_3	1	0	1	1	1	1			

(a) Find Need Matrix.

(b) Is the system in a safe state? Why or why not?

III. What do you mean by IPC? Why is it required? Explain the two fundamental models of it.

IV. Explain Dining Philosopher problem with its semaphore solution and specify how it meets all three requirements of solution to its critical section problem?

Printed Pages: 02 University Roll No

Second Term Examination, 2014-15
Course: B.Tech. (CSE), Year: II , Semester: IV
Database Management System (ITE 221)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Why certain functional dependencies are called trivial functional dependencies?

II. What are Full and Partial functional dependencies?

III. Which normal form is considered adequate for normal relational database design?

IV. Consider a schema R (A, B, C, D) and functional dependencies A→ B and C → D. Show that the decomposition of R into R1 (A, B) and R2(C, D) is Loss less join or Lossy join?

V. Define the term functional dependencies.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. List the three design goals for relational databases, and explain why each is desirable.

II. Compute the closure of the following set F of functional dependencies for relation Schema R = (A, B, C, D, E).
A →BC, CD →E, B→ D, E→ A . List the candidate keys for R.

III. A table has fields F1, F2, F3, F4, and F5, with the following functional dependencies:

F1→F3, F2→F4, (F1, F2) →F5.

In terms of normalization, this table exists in which normal form justify it and further normalize it.

IV. What are the different type of anomalies explain each with suitable example?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Consider the following relational database. Give an expression in SQL for each of the following queries

employee (employee-name, street, city)
works (employee-name, company-name, salary)
company (company-name, city)
manages (employee-name, manager-name)

a. Find the names of all employees who work for First Bank Corporation.

b. Find the names and cities of residence of all employees who work for First Bank Corporation.

c. Find the names, street addresses, and cities of residence of all employees who work for First Bank Corporation and earn more than $10,000.

d. Find all employees in the database who live in the same cities as the companies for which they work.

e. Find the company that has the smallest payroll. (1)

II. In a relation schema R(ABCDEH), if F = {A→BCD, CD→E, E→CD, D→AH, ABH→BD, DH→BC} then find out canonical Cover of F.

III. Why normalization is required? Explain 1NF, 2NF, 3NF normal form with a suitable example.

IV. Write a short note on the following:
(a) Cursor (b) Trigger (c)Views

Mid Term Examination,
Even Semester 2014-15

Course: B. Tech. II Yr.
Title of Paper: Environmental Studies
Paper Code: CEE-201
Time: 1.5 Hours

Univ. No. ----------

Max. Marks: 20

Section-A

Note: Attempt All Questions

1 X 5 = 5 Marks

(I) Define stone leprosy?
(II) What is salinisation?
(III) Define ex-situ conservation with example.
(IV) Write names of four important GHGs.
(V) What is age structure?

Section-B

Note: Attempt Any three Questions

2 x 3 = 6 Marks

(I) "Hydrogen is an alternative source of future energy" comment on it.
(II) Write differences between conventional and non-conventional sources of energy.
(III) Write types and sources of solid wastes.
(IV) How ozone is formed in stratosphere?

Section-C

Note: Attempt Any Three Questions

3 x 3 = 9 Marks

(I) What is solar energy? Discuss advantages and limitations of solar energy.
(II) What are the effects of transportation on environment.
(III) Define air pollution. Explain its sources and effects.
(IV) Discuss the cause, effects and control of measure of global warming.